Q Pathformer v08 Priority Training and Routed Models

Internal findings report for Anthony Siljeg

QPATH RPT009 • 21 October 2026 • Internal research • Classical computation only

Decision and measured findings

Priority sampling produced small, task-dependent benefits in this controlled Q-Pathformer study. It improved the registered probability-loss comparison for the routed trigonometric model on checkerboard, spiral and rare disks, and for the routed circuit on rare disks. The other four direct priority comparisons were inconclusive. None established an improvement exceeding the predeclared 0.01 BCE practical reference across its whole interval.

The useful result is a bounded interaction between how examples are sampled and how the model routes them. On checkerboard and spiral, the trigonometric model benefited more from priority with geometric routing than with hash routing. Priority did not close the routed circuit's large loss gap against the wider global classical control on either task. The next research emphasis should be representation and routing, with priority retained as an optional policy.

Task	Direct priority finding	Remaining limitation
Checkerboard	TrigK improves BCE	Circuit still substantially behind global control
Spiral	TrigK improves BCE	Same model-class gap; native loss shows extreme errors
Noisy annulus	Both direct effects inconclusive	Priority revisits flipped labels; no confirmed primary gain
Rare disks	MultiK and TrigK improve BCE	Small effects; highest F1 does not identify best probability loss

All ten arms and all twenty registered primary comparisons are reported. Macro-F1 is descriptive; the primary metric is common clipped binary cross-entropy (BCE), where lower is better. Confirmation contains eight independent populations per task, with two paired optimizer seeds inside each. The study evaluates a small classical emulation of two-qubit circuits and classical controls. It supplies no physical quantum result or quantum-advantage claim.

Classical inference findings

Exact classical conversion is the clearest deployment result. Across the four tasks and four circuit-policy arms, the reported group medians of paired original-to-compiled latency ratios range from 4.33 to 4.59 for single inputs, 11.82 to 16.03 for batches of 32, and 2.33 to 5.36 for batches of 1,024. Each ratio compares complete calls with resident model data on this CPU. Independent prediction checks found a maximum probability difference below 2.62e-14 and no classification changes. These are faster implementations of the same fitted predictors, not improvements in F1 or measurements of quantum hardware.

The speed benefit has limits. The compiled circuit uses 1,000 coefficient bytes instead of 560 original parameter bytes, or 1,560 bytes if both are retained, before router and container overhead. Strong learned classical controls still provide lower latency in several conditions and better predictive performance on the hardest tasks. Session three was substantially faster than the first two; all three sessions remain in the reported summaries. These descriptive measurements support a useful classical deployment option while leaving hardware-independent speed, cold-start cost and broader architecture claims unestablished.

Research question and historical context

This study isolates a training-policy question within a small Q-Pathformer implementation: can giving difficult training examples more sampling opportunities improve a fixed routed predictor, and does the effect depend on meaningful geometric routing? Each retained model is paired with a uniform and a priority sampling policy. Model capacity, data splits, full update budgets and checkpoint selection rules are fixed before confirmation testing.

The circuit models use classical numerical emulation of two qubits. Five separate parameter banks are available, and the frozen router selects the bank used for an input. The experiment does not test a large language model, a quantum processor, learned semantic priorities, retention policies, online adaptation or the full proposed architecture. Q-Path Multiplier (ETEL) schedules complete fits; its logical slots do not create additional scientific states or model capacity.

The sealed v0.7 study motivated this intervention by finding that geometric routing outperformed the particular hash routing on the four synthetic families, while a wider global trigonometric control remained much stronger on the difficult checkerboard and spiral patterns. v0.8 keeps the bounded predictor families and tests a training-only priority policy with new populations and fixed controls.

Historical scores are context, not a pooled experiment or a direct improvement measure. v0.7 used shuffled record passes; v0.8 uses draws with replacement for both policies. The fresh population, optimizer and routing variation is evaluated within v0.8. Earlier high scores on simpler tasks or text classification do not establish broad architecture performance, language-model capability or superiority over modern AI systems.

Study design and hardware

MultiK and MultiHash share the same two-layer, two-qubit circuit family. Each layer encodes the two input coordinates with RY rotations, applies learned RY and RZ rotations, then a controlled-X operation. A learned affine head combines X0, X1, Z0, Z1 and Z0Z1 expectation values, followed by a

sigmoid. Five banks contain 70 learned predictor parameters in total. The geometric and hash versions differ in the frozen routing rule.

TrigK and TrigHash use five independent 25-coefficient real trigonometric predictors, giving 125 learned parameters. TrigG121 uses one global 121-coefficient trigonometric predictor with a wider frequency basis. Counts exclude router metadata, optimizer state and retained audit traces. Similar coefficient counts do not establish equal function classes, equal optimization difficulty or equal computational cost.

The geometric router is fitted only on training coordinates using five k-means centers and the registered deterministic restart procedure. The hash router uses a salted coordinate digest and four frozen cutpoints chosen to match the geometric router's training-bank counts exactly. Held-out occupancy can differ. Labels, split identifiers and model errors do not enter the routing rule; priority sampling changes exposure to existing records without refitting the router.

Base model	Router	Learned predictor parameters	Representation
MultiK	Five kmeans banks	70	Two qubits and learned five-moment head
MultiHash	Five hash banks	70	Same circuit and head
TrigK	Five kmeans banks	125	Five 25-coefficient predictors
TrigHash	Five hash banks	125	Same local trigonometric basis
TrigG121	Global	121	Wider global frequency basis

Each base has a uniform arm U and a priority arm P. Router coordinates or cutpoints, Adam state, cached features and audit traces are additional storage. The ten centroid coordinates and the hash salt plus four cutpoints are separate from predictor parameter counts.

The four synthetic task families are checkerboard, spiral, noisy annulus and rare disks. Each population has 8,000 independently generated records, split into 4,800 training, 1,600 validation and 1,600 test records using observed-label stratification. The annulus has independent 10% observed-label flips. Hidden clean labels and flip indicators are isolated from the trainer and become available only for supplementary diagnostics after the test release.

The registered matrix contains 40 short engineering fits, 80 development fits and 640 confirmation fits. Confirmation uses eight fresh populations per task and two paired optimizer seeds within each population. Full fits run 200 rounds of 75 minibatch updates, or 15,000 updates with batch size 64. Engineering uses 12 rounds and 900 updates. Four additional preselected clean-directory retrains are reproducibility checks and do not increase the independent sample size.

Adam uses learning rate 0.01, betas 0.9 and 0.999, and epsilon 1e-8, with no decay, scheduler or gradient clipping. Every arm completes its full horizon. The selected checkpoint is the earliest exact minimum of full-validation clipped binary cross-entropy over that horizon. Neither selection nor the sampler uses test outcomes, and weak development results do not remove an arm.

Audit coverage is explicit. The author auditor recalculates first and last minibatch gradients in every round. The independent auditor recalculates first and last minibatches in rounds 1, 10, 11, 12, 100 and 200, plus the selected full-training gradient. All retained gradients drive an independent Adam recurrence, while independent forward calculations check every end-round training and validation

prediction and every priority refresh. Recurrence from stored gradients does not independently prove every unprobed gradient correct.

Task definitions

Inputs are two real coordinates drawn uniformly from $[-1,1]^2$. Class one is defined by the following conditions; observed annulus labels additionally undergo independent Bernoulli 0.10 flips. The other families use the clean deterministic label.

Task	Class one condition
Checkerboard	$\sin(3\pi x_0 + \pi/7)\sin(3\pi x_1 + \pi/5) > 0$
Spiral	$\sin(2\,\mathrm{atan2}(x_1,x_0) + 4\pi\sqrt{(x_0^2+x_1^2)}) > 0$
Noisy annulus	$0.25 < x_0^2+x_1^2 < 0.85$ before label flips
Rare disks	Squared distance to the nearest of (−0.55,−0.45), (0.5,−0.35), (0,0.55) is below 0.22^2

Freshness checks cover 320,000 accepted new coordinates and exclude all available prior coordinates, including 584,000 official historical records and 67,222 retained fixture coordinates. There were no redraws. This verifies exact canonical-coordinate separation, not near-duplicate exclusion or real-world novelty. The matrix records 548 distinct 128-bit RNG streams, disjoint from the historical seed inventory.

The recorded host is an MSI Stealth A18 AI+ A3XWJG with an AMD Ryzen AI 9 HX 370 processor, 12 physical cores and 24 logical processors. Windows reports 67,772,469,248 bytes of physical memory (approximately 63.1 GiB). The operating system is Windows 11 Pro, version 10.0.26300, build 26300. The study uses the CPU; the GPU is not used.

The preserved scientific environment is Python 3.12.14, NumPy 2.5.4 and SciPy 1.18.1. Effective OpenBLAS threading is checked at one thread per scientific process. ETEL admits at most five complete fits concurrently. Whole-process CPU, wall time, sampler and refresh costs, audit work and storage are separate accounting scopes. A five-worker observation does not establish optimal concurrency or a speedup over another worker count.

Whole-worker CPU, phase wall and nested sampler, refresh and full-training diagnostic timers have different scopes and must not be added together. Training wall includes validation and evidence writes; its timers end before final result and commit serialization. Pure gradient-computation and filesystem-I/O durations are not isolated by the trainer. Repeated metadata charges for original data preparation are not repeated host work. Audit CPU belongs to the actual audit interpreter even where an inherited helper label says fit child.

Priority sampling and comparison fairness

Both policies use sampling with replacement. The first ten rounds are uniform. At the start of each later round, native logit binary cross-entropy is calculated for every training record. The score is the square root of that native loss capped at five plus 0.001 inside the square root. The priority probability is one half the uniform probability plus one half the record score divided by the sum of all training scores. The distribution stays fixed for that round. The priority policy draws from this distribution; the uniform control retains the proposal only as a diagnostic and continues to draw uniformly.

Each sampled data gradient is multiplied by one divided by the product of the number of training records and that record's actual sampling probability, before the minibatch mean. After warmup, the actual probability is the priority probability for the priority policy; it is uniform for the uniform policy and all warmup rounds, so those weights are one. The diagnostic priority proposal never weights a uniform-control gradient. Weights are neither normalized nor clipped, and the sampler is not differentiated. The uniform mixture bounds the weight at two. Conditional on the sampling distribution, the corrected raw gradient targets ordinary empirical binary cross-entropy; this does not imply identical Adam trajectories or guaranteed generalization gains.

Within each base model, paired uniform and priority arms share their initial parameters and first ten training states. All arms within a population and seed group share the underlying uniform random draws; this does not imply identical parameters across different model families. Uniform controls also perform the scheduled diagnostic loss refreshes. This padding helps compare the registered policies under matched instrumentation but cannot establish the native end-to-end cost of an uninstrumented uniform trainer. Any subtraction of refresh cost must be labelled an estimate.

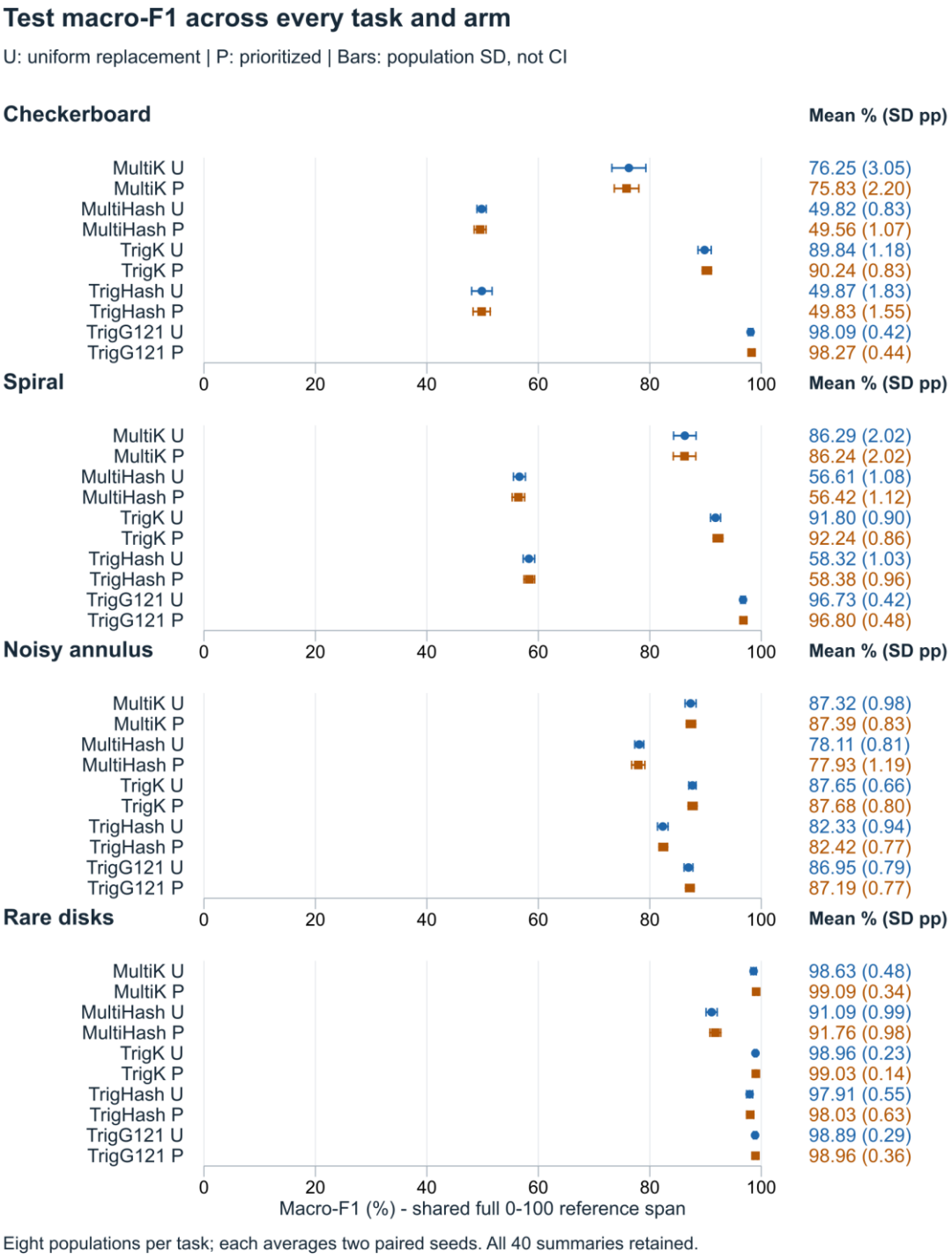
The primary outcome is common clipped natural-log binary cross-entropy. Five predeclared contrasts are evaluated separately on each of four tasks: the priority effect in MultiK, the priority effect in TrigK, the corresponding two geometric-versus-hash interactions, and MultiK_P versus TrigG121_P. The global trigonometric priority effect is secondary.

For each contrast, the two paired seed differences are averaged within each population. Inference uses eight independent population means per task, not sixteen independent runs. The registered two-sided t interval has seven degrees of freedom and 99.75% nominal per-contrast confidence for the twenty-comparison Bonferroni family. Small-sample distributional assumptions remain material. Crossing zero is inconclusive rather than equivalence, and a favorable interaction need not mean either priority arm improves in absolute terms.

Macro-F1, accuracy, raw Brier score, fixed-bin calibration, native loss, positive-class precision and recall, average precision, bank exposure and boundary slices are secondary or descriptive. Validation milestones at 90%, 95% and 97% macro-F1 retain nonreachers explicitly; 97% is not an expectation for the noisy annulus. No sequence of test checkpoints is used to select a favorable stopping point.

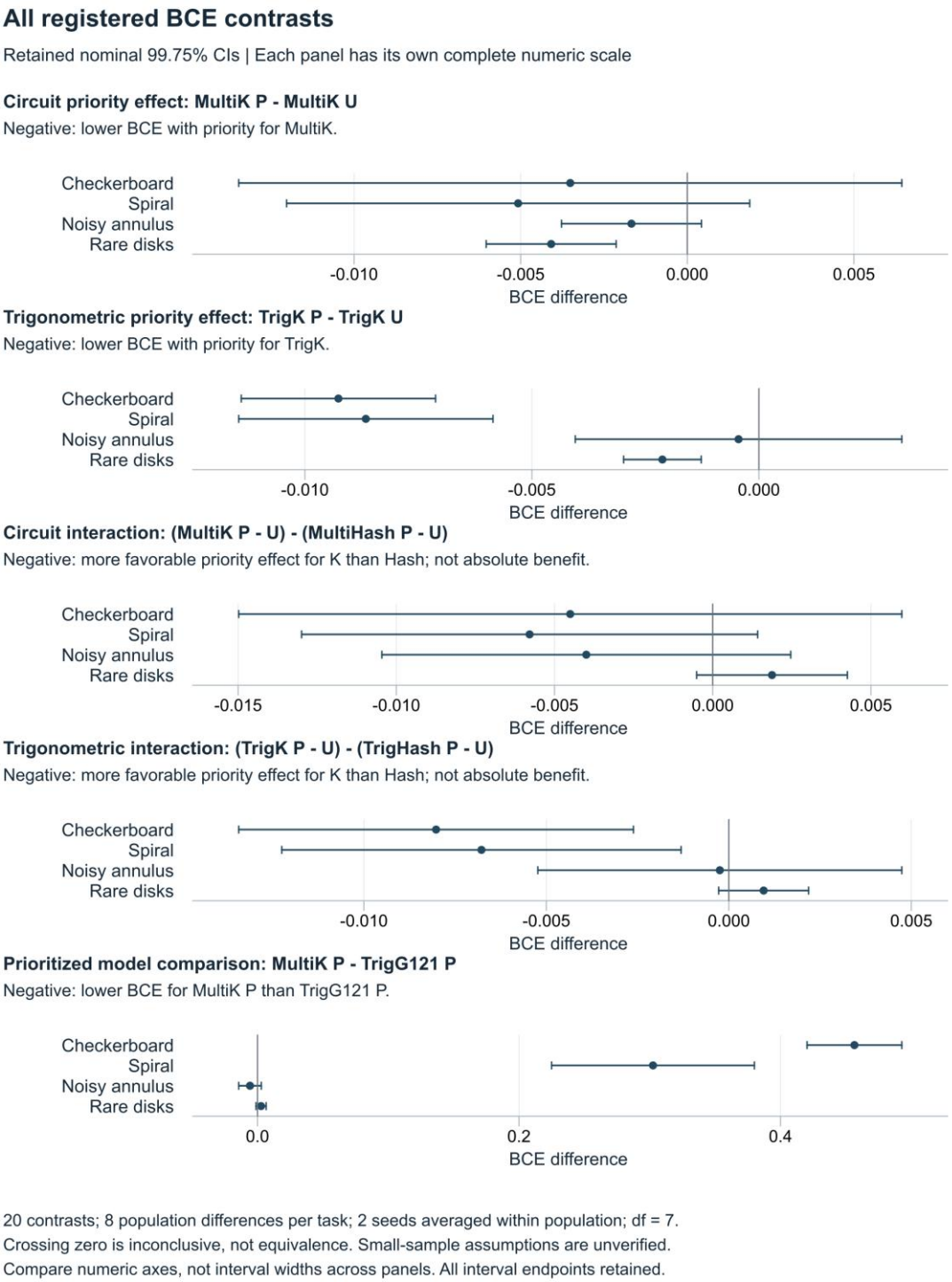
The controlled intervention begins after 750 uniform updates. All full fits still receive 960,000 sampled record visits and 15,000 updates. Loss-based priority is not a semantic rating of information quality, a retention policy or a learned routing mechanism. The probabilities use training labels only; validation selects a checkpoint after the complete horizon, and test outcomes were released only after every confirmation endpoint passed both audits.

All task macro F1 summaries



All 40 task/arm summaries. Points are retained mean test macro-F1; whiskers are ± one retained population sample SD, not confidence intervals. Each of eight population values averages two paired training seeds. U = uniform replacement sampling; P = prioritized sampling. Adjacent U/P rows retain the matched policy comparison within each base model; no ranking or outcome filtering is applied. All panels share an axis containing the full 0–100% reference span, expanded only if an SD whisker extends beyond it; SD whiskers are not clipped to the metric bounds. Right-hand text is mean (population SD): means are percentages and SD values are percentage points. Higher F1 is better. These are classical results on four synthetic task families.

All primary BCE comparisons



All 20 preregistered common-BCE contrasts. Points and nominal 99.75% Bonferroni t confidence intervals are copied from the committed analysis (20 comparisons, eight independent population differences, df = 7); no interval is recalculated. Two paired-seed differences are averaged within each population. Each contrast panel has its own clearly labelled numeric x-axis including zero and every complete interval; widths cannot be compared across panels. Differences use the displayed first-minus-second expression. Negative direct priority effects mean lower BCE with P than U; a negative K-minus-Hash interaction means a more favorable priority effect for K relative to Hash and does not alone show an absolute benefit. For MultiK P minus TrigG121 P, negative favors MultiK P. Crossing zero is inconclusive, not equivalence. Small-sample assumptions remain unverified. No physical quantum result or hardware advantage is implied.

Routing boundaries noise and imbalance

Geometric models have better descriptive mean BCE and F1 than their corresponding hash models on all four task families under both policies. This consistent pattern supports continuing geometry-aware routing research, but the absolute K-versus-Hash comparison was not added as a new primary test in v0.8. Matching training-bank occupancy controls one allocation feature; it does not match spatial continuity, held-out occupancy or effective function class.

All models are scored on the same frozen geometric-margin slices. Margin is the second-nearest minus nearest squared centroid distance. Training quantile edges at 0, 0.1, 0.25, 0.5 and 1 are fixed before held-out scoring; duplicate edges are collapsed deterministically. Router-specific bank indices and margins vary between populations, so slice comparisons are descriptive. The evidence retains all 560 task-arm-policy-bin summaries, with counts, class prevalence and population SD/range. All 4,480 population-slice blocks were observed; the implementation would retain an empty-bin status if one occurred.

Thirty retained slice summaries contain at least one single-class population block. For a slice with only one true class, the fixed two-class macro-F1 convention can score at most 0.5 even when the present class is predicted perfectly. This must be read alongside class counts, BCE and accuracy, rather than interpreted automatically as a failed model. Undefined ratios and empty slices are retained explicitly.

On annulus, flipped labels are 9.93% of training records on average. Uniform draws visit them at approximately 9.92%; priority draw fractions rise to 18.46% for MultiK, 14.09% for MultiHash, 19.05% for TrigK, 16.28% for TrigHash and 19.30% for TrigG121. These flags were inspected only after test release and were never model or sampler inputs. More raw visits do not imply more conditional expected raw-gradient weight under importance correction. The result does not prove label-noise overfitting or a noise-detection capability.

Rare-disk precision, recall and grouped-tie average precision are essential. MultiHash_U reaches 96.54% accuracy yet only 91.09% F1 and 81.91% positive recall. MultiHash_P remains weaker at 91.76% F1 and 83.13% recall. MultiK_P reaches 98.43% positive recall, while the global control has lower descriptive probability loss. Neither a high accuracy score nor the single highest F1 establishes overall model superiority.

Per-bank gradient norms, minibatch appearances, learned parameter norms and packed-Adam motion during absent-bank updates are retained. An absent data gradient does not freeze its parameters when Adam carries momentum from earlier steps. These observations describe the implemented optimizer; they do not identify a causal advantage of state retention or validate an online architecture.

Clean reproduction and classical deployment

Four preselected same-environment runs were regenerated and fitted in a fresh directory, using 200 rounds and 15,000 optimizer updates each. No original checkpoint was copied. The independent comparison covers complete training and validation curves, retained sampler traces, selected and final checkpoints, and exact selections/classifications. These four checks add no independent populations and are not a portability or cross-hardware replication claim.

Task and arm	Selected round	Max parameter error	Max logit error	Max curve metric error
Checkerboard MultiK P	183	0.000e+00	0.000e+00	5.551e-17
Spiral TrigK P	199	0.000e+00	0.000e+00	5.551e-17
Noisy annulus MultiHash U	128	0.000e+00	0.000e+00	4.163e-17
Rare disks TrigG121 U	186	0.000e+00	0.000e+00	5.551e-17

All four registered checks retained. Reported maxima are copied/composed from independent comparison records; exact classifications compared and all field-level errors remain in DERIVED_EVIDENCE.json. Same-environment tolerances are 1e-12 for parameters/metrics and 1e-10 for native logits; selected steps and classes are exact.

Exact compiled representations

The conversion phase retains all 256 circuit coefficient tables and all 384 learned-trigonometric same-checkpoint repacks. Independent Laurent/dense calculations and separate feature expansion check all 640 models, validation and test predictions, and label-free off-grid and routing-boundary fixtures. Repacking the learned controls provides the same optimized feature kernel. Conversion changes representation and execution cost; it does not add training or an accuracy improvement.

Independent comparison	Maximum absolute error	Recorded comparisons
maximum probability error	2.615e-14	1280
maximum native logit error	1.057e-13	1280
coefficient max abs error	2.132e-14	1792
converted dense logit max abs error	8.527e-14	1792
saved probability max abs error	9.437e-15	1792
saved logit max abs error	7.816e-14	1792
probability error	2.165e-14	384
native logit error	1.019e-13	384

The independent records contain 0 threshold-crossing entries across 11008 explicit comparison records. These entries may overlap across reference comparisons or fixtures. Every accepted crossing must have both probabilities within the frozen 32-epsilon band around 0.5; this is a floating-point qualification, not permission for arbitrary class changes. Full locations and values remain retained.

Arm	Original bytes	Coefficient bytes	NPZ bytes	Build median µs
MultiK U	560–560	1000–1000	10588–10588	1281.40
MultiK P	560–560	1000–1000	10588–10588	1131.85
MultiHash U	560–560	1000–1000	10588–10588	1110.40
MultiHash P	560–560	1000–1000	10588–10588	1201.10
TrigK U	1000–1000	1000–1000	1278–1278	16.80
TrigK P	1000–1000	1000–1000	1278–1278	17.10
TrigHash U	1000–1000	1000–1000	1278–1278	16.25
TrigHash P	1000–1000	1000–1000	1278–1278	16.30
TrigG121 U	968–968	968–968	1246–1246	16.75
TrigG121 P	968–968	968–968	1246–1246	17.15

All 64 models per arm across the four tasks are included. Bytes show complete ranges; construction is the median of 64 single instrumented transforms, not a repeated isolated benchmark or inferential task comparison. Float64 parameter payload, coefficient payload and representation-container storage are separate. Full 640-model construction/load/storage records are retained; router, filesystem and retained-parameter overhead are not hidden in the coefficient count.

Compilation is not compression. The circuit's original float64 parameters occupy 560 bytes; its coefficient table occupies 1,000 bytes, and retaining both requires 1,560 bytes before router, container and other overhead. The benefit being tested is execution cost while preserving predictions.

Limitations and next research step

The results support a narrow contribution to the v1.0 paper: a reproducible, controlled study of loss-based priority, fixed routing and small classical two-qubit emulation. Geometric routing matters on these synthetic patterns, and priority has selective modest benefits. The circuit family is not generally the most accurate predictor, and it remains substantially weaker than the global trigonometric control on checkerboard and spiral.

Representation, routing geometry and optimization remain entangled in the model-class comparison. Similar parameter counts do not equalize expressive functions, spatial smoothness, optimization landscape or compute. Eight synthetic populations per task improve estimation of same-family data/router variation but do not establish generalization to real-world or out-of-distribution inputs. Two nested seeds and technical timing repeats do not add independent task populations.

The full proposed architecture remains largely untested: large numbers of states, learned semantic routing, long sequences, language-model training, online state retention, distributed elasticity and scaling beyond two emulated qubits were outside this study. Priority uses current training loss, not a semantic assessment of data value. Exact classical conversion exploits this small circuit's mathematical structure and supplies no evidence for physical quantum efficiency.

For a separately approved v0.9 plan, retain uniform and priority variants, geometric and hash controls, and the strongest global classical baseline on a fresh, modest real-data classification benchmark. Freeze the preprocessing, validation rule, capacity and cost controls before testing. Focus on whether useful routing generalizes and whether representation restrictions or optimization explain the present gaps. Do not retroactively tune priority, use hidden noise flags or add favorable tests to v0.8.

Keep priority optional until its task-specific benefit and unpadded operational cost are tested. A noise-robust sampler, an early-stopping policy or a portable installation benchmark each needs a separate registered comparison. Present v1.0 as classical-only evidence with reproducible methods and limitations. Physical quantum execution and a matched classical-versus-quantum comparison remain planned v1.1 work.

Appendix A Complete predictive performance

Every entry below averages two seeds within each of eight independent populations, then reports the population mean and sample SD. F1 and accuracy are percentages; their SD is in percentage points. BCE, Brier and ECE are on their native scales. These are descriptive variations, not confidence intervals. All arms use the validation-selected checkpoint from the full training horizon.

These tables retain native, unclipped logit BCE and positive-class metrics for every arm. Values are mean and population SD. Precision, recall and grouped-tie average precision are percentages. Primary inference continues to use the registered clipped BCE; native loss exposes extreme errors that clipping can suppress. No undefined population ratio is replaced with zero.

Checkerboard

Priority slightly reduces circuit loss while its mean F1 decreases from 76.25% to 75.83%. TrigK improves to 90.24% F1, while the global control reaches 98.27%. Both hash bases remain near chance. A modest ECE for a near-chance predictor does not make it useful.

Arm	BCE	Macro F1 %	Accuracy %	Brier	ECE10
MultiK U	0.5062 ± 0.0215	76.25 ± 3.05	76.27 ± 3.05	0.1670 ± 0.0091	0.0697 ± 0.0202
MultiK P	0.5027 ± 0.0212	75.83 ± 2.20	75.86 ± 2.20	0.1662 ± 0.0085	0.0631 ± 0.0144
MultiHash U	0.6997 ± 0.0023	49.82 ± 0.83	50.02 ± 0.93	0.2532 ± 0.0011	0.0480 ± 0.0087
MultiHash P	0.7007 ± 0.0029	49.56 ± 1.07	49.71 ± 1.07	0.2537 ± 0.0014	0.0504 ± 0.0100
TrigK U	0.2688 ± 0.0102	89.84 ± 1.18	89.84 ± 1.18	0.0800 ± 0.0045	0.0716 ± 0.0107
TrigK P	0.2595 ± 0.0106	90.24 ± 0.83	90.25 ± 0.83	0.0772 ± 0.0044	0.0696 ± 0.0089
TrigHash U	0.7035 ± 0.0060	49.87 ± 1.83	50.04 ± 1.91	0.2550 ± 0.0029	0.0580 ± 0.0163
TrigHash P	0.7023 ± 0.0055	49.83 ± 1.55	49.97 ± 1.54	0.2544 ± 0.0027	0.0565 ± 0.0169
TrigG121 U	0.0530 ± 0.0059	98.09 ± 0.42	98.09 ± 0.42	0.0153 ± 0.0022	0.0188 ± 0.0010
TrigG121 P	0.0461 ± 0.0057	98.27 ± 0.44	98.27 ± 0.44	0.0135 ± 0.0021	0.0161 ± 0.0019

Checkerboard observed test performance. Mean and population SD; n = 8 populations per arm.

Arm	Native BCE	Precision %	Recall %	Average precision %
MultiK U	0.5062 ± 0.0215	76.37 ± 2.90	76.45 ± 3.88	83.33 ± 2.46

Arm	Native BCE	Precision %	Recall %	Average precision %
MultiK P	0.5027 ± 0.0212	76.07 ± 2.56	75.78 ± 2.84	83.38 ± 2.36
MultiHash U	0.6997 ± 0.0023	50.06 ± 1.10	50.07 ± 5.88	50.77 ± 0.99
MultiHash P	0.7007 ± 0.0029	49.79 ± 1.11	49.26 ± 4.31	50.08 ± 1.29
TrigK U	0.2688 ± 0.0102	89.70 ± 1.08	90.08 ± 1.45	96.54 ± 0.60
TrigK P	0.2595 ± 0.0106	89.93 ± 0.77	90.70 ± 1.45	96.83 ± 0.56
TrigHash U	0.7035 ± 0.0060	50.07 ± 1.75	50.13 ± 6.03	50.16 ± 1.20
TrigHash P	0.7023 ± 0.0055	50.05 ± 1.29	48.81 ± 4.58	50.35 ± 1.37
TrigG121 U	0.0530 ± 0.0059	98.37 ± 0.45	97.83 ± 0.73	99.89 ± 0.04
TrigG121 P	0.0461 ± 0.0057	98.46 ± 0.47	98.08 ± 0.73	99.91 ± 0.04

Spiral

TrigK priority raises mean F1 from 91.80% to 92.24%; the wider global control is approximately 96.80%. Circuit priority changes mean F1 from 86.29% to 86.24%. Its loss gap to the global control remains supported by the primary comparison.

Arm	BCE	Macro F1 %	Accuracy %	Brier	ECE10
MultiK U	0.3879 ± 0.0416	86.29 ± 2.02	86.30 ± 2.01	0.1104 ± 0.0157	0.0759 ± 0.0187
MultiK P	0.3828 ± 0.0432	86.24 ± 2.02	86.25 ± 2.01	0.1091 ± 0.0164	0.0708 ± 0.0163
MultiHash U	0.6703 ± 0.0067	56.61 ± 1.08	56.71 ± 1.14	0.2395 ± 0.0031	0.0463 ± 0.0098
MultiHash P	0.6710 ± 0.0061	56.42 ± 1.12	56.49 ± 1.10	0.2398 ± 0.0029	0.0472 ± 0.0087
TrigK U	0.2490 ± 0.0281	91.80 ± 0.90	91.80 ± 0.89	0.0649 ± 0.0082	0.0488 ± 0.0118
TrigK P	0.2404 ± 0.0277	92.24 ± 0.86	92.25 ± 0.86	0.0621 ± 0.0076	0.0467 ± 0.0124
TrigHash U	0.6685 ± 0.0060	58.32 ± 1.03	58.37 ± 1.07	0.2383 ± 0.0027	0.0498 ± 0.0094

Arm	BCE	Macro F1 %	Accuracy %	Brier	ECE10
TrigHash P	0.6666 ± 0.0069	58.38 ± 0.96	58.43 ± 0.99	0.2375 ± 0.0030	0.0497 ± 0.0086
TrigG121 U	0.0810 ± 0.0088	96.73 ± 0.42	96.73 ± 0.42	0.0237 ± 0.0023	0.0097 ± 0.0023
TrigG121 P	0.0803 ± 0.0103	96.80 ± 0.48	96.80 ± 0.48	0.0233 ± 0.0028	0.0110 ± 0.0022

Spiral observed test performance. Mean and population SD; n = 8 populations per arm.

Arm	Native BCE	Precision %	Recall %	Average precision %
MultiK U	0.3890 ± 0.0407	86.88 ± 2.43	85.40 ± 2.06	92.19 ± 1.82
MultiK P	0.3843 ± 0.0419	86.98 ± 2.86	85.19 ± 2.50	92.40 ± 1.91
MultiHash U	0.6703 ± 0.0067	56.71 ± 1.20	55.24 ± 3.14	62.10 ± 1.23
MultiHash P	0.6710 ± 0.0061	56.43 ± 1.11	55.19 ± 2.62	61.95 ± 1.01
TrigK U	0.2575 ± 0.0225	92.27 ± 0.74	91.17 ± 1.53	96.94 ± 0.50
TrigK P	0.2510 ± 0.0226	92.72 ± 1.26	91.64 ± 1.28	97.17 ± 0.50
TrigHash U	0.6685 ± 0.0060	58.43 ± 0.99	56.72 ± 1.59	62.74 ± 0.63
TrigHash P	0.6666 ± 0.0069	58.50 ± 0.98	56.85 ± 0.99	62.92 ± 0.68
TrigG121 U	0.0810 ± 0.0088	96.76 ± 0.74	96.68 ± 0.36	99.62 ± 0.06
TrigG121 P	0.0803 ± 0.0103	96.93 ± 0.92	96.63 ± 0.31	99.63 ± 0.07

Noisy annulus

All five primary comparisons on noisy annulus are inconclusive. The kmeans and global means are close in the observed-label metric, around 87% F1, with no registered basis for declaring priority generally beneficial. Hash controls remain weaker.

Arm	BCE	Macro F1 %	Accuracy %	Brier	ECE10
MultiK U	0.4002 ± 0.0203	87.32 ± 0.98	87.37 ± 0.99	0.1149 ± 0.0067	0.0771 ± 0.0087
MultiK P	0.3985 ± 0.0200	87.39 ± 0.83	87.43 ± 0.84	0.1146 ± 0.0063	0.0768 ± 0.0081

Arm	BCE	Macro F1 %	Accuracy %	Brier	ECE10
MultiHash U	0.5185 ± 0.0100	78.11 ± 0.81	78.13 ± 0.80	0.1645 ± 0.0037	0.0679 ± 0.0072
MultiHash P	0.5208 ± 0.0112	77.93 ± 1.19	77.95 ± 1.17	0.1652 ± 0.0045	0.0669 ± 0.0081
TrigK U	0.3954 ± 0.0159	87.65 ± 0.66	87.70 ± 0.67	0.1130 ± 0.0057	0.0773 ± 0.0034
TrigK P	0.3949 ± 0.0173	87.68 ± 0.80	87.73 ± 0.81	0.1130 ± 0.0058	0.0759 ± 0.0057
TrigHash U	0.4659 ± 0.0199	82.33 ± 0.94	82.34 ± 0.94	0.1397 ± 0.0070	0.0631 ± 0.0071
TrigHash P	0.4657 ± 0.0202	82.42 ± 0.77	82.43 ± 0.78	0.1402 ± 0.0068	0.0659 ± 0.0054
TrigG121 U	0.4059 ± 0.0218	86.95 ± 0.79	87.00 ± 0.80	0.1144 ± 0.0070	0.0740 ± 0.0061
TrigG121 P	0.4042 ± 0.0216	87.19 ± 0.77	87.24 ± 0.78	0.1139 ± 0.0066	0.0732 ± 0.0078

Noisy annulus observed test performance. Mean and population SD; n = 8 populations per arm.

Arm	Native BCE	Precision %	Recall %	Average precision %
MultiK U	0.4002 ± 0.0203	87.56 ± 1.31	85.75 ± 1.27	86.23 ± 1.62
MultiK P	0.3985 ± 0.0200	87.57 ± 1.06	85.88 ± 1.43	86.14 ± 1.62
MultiHash U	0.5185 ± 0.0100	74.16 ± 0.93	83.21 ± 1.49	78.56 ± 1.53
MultiHash P	0.5208 ± 0.0112	74.16 ± 1.09	82.63 ± 1.13	78.48 ± 1.52
TrigK U	0.3954 ± 0.0159	88.40 ± 0.88	85.49 ± 0.83	86.06 ± 1.39
TrigK P	0.3949 ± 0.0173	88.61 ± 1.21	85.30 ± 1.08	86.13 ± 1.40
TrigHash U	0.4659 ± 0.0199	79.18 ± 0.92	85.52 ± 1.16	83.83 ± 1.09
TrigHash P	0.4657 ± 0.0202	79.23 ± 0.66	85.68 ± 1.13	83.75 ± 1.23
TrigG121 U	0.4059 ± 0.0218	87.66 ± 1.02	84.71 ± 1.14	86.09 ± 1.72
TrigG121 P	0.4042 ± 0.0216	87.90 ± 0.98	85.00 ± 1.10	86.11 ± 1.89

Rare disks

MultiK_P has the highest descriptive F1 mean, 99.09%, but TrigG121_P has lower mean BCE. Their registered BCE comparison is inconclusive. MultiHash accuracy appears high while its positive recall remains substantially lower; the secondary tables retain this imbalance.

Arm	BCE	Macro F1 %	Accuracy %	Brier	ECE10
MultiK U	0.0158 ± 0.0035	98.63 ± 0.48	99.46 ± 0.19	0.0043 ± 0.0011	0.0066 ± 0.0011
MultiK P	0.0117 ± 0.0031	99.09 ± 0.34	99.64 ± 0.14	0.0032 ± 0.0008	0.0050 ± 0.0007
MultiHash U	0.0875 ± 0.0087	91.09 ± 0.99	96.54 ± 0.40	0.0255 ± 0.0027	0.0137 ± 0.0019
MultiHash P	0.0816 ± 0.0096	91.76 ± 0.98	96.80 ± 0.38	0.0238 ± 0.0028	0.0128 ± 0.0016
TrigK U	0.0127 ± 0.0018	98.96 ± 0.23	99.59 ± 0.09	0.0035 ± 0.0005	0.0056 ± 0.0012
TrigK P	0.0106 ± 0.0018	99.03 ± 0.14	99.61 ± 0.06	0.0030 ± 0.0006	0.0042 ± 0.0007
TrigHash U	0.0237 ± 0.0035	97.91 ± 0.55	99.17 ± 0.22	0.0066 ± 0.0012	0.0089 ± 0.0007
TrigHash P	0.0206 ± 0.0040	98.03 ± 0.63	99.21 ± 0.25	0.0060 ± 0.0014	0.0061 ± 0.0011
TrigG121 U	0.0095 ± 0.0024	98.89 ± 0.29	99.56 ± 0.12	0.0029 ± 0.0009	0.0035 ± 0.0005
TrigG121 P	0.0090 ± 0.0027	98.96 ± 0.36	99.59 ± 0.15	0.0028 ± 0.0009	0.0032 ± 0.0006

Rare disks observed test performance. Mean and population SD; n = 8 populations per arm.

Arm	Native BCE	Precision %	Recall %	Average precision %
MultiK U	0.0158 ± 0.0035	97.79 ± 0.84	97.35 ± 1.66	99.80 ± 0.15
MultiK P	0.0117 ± 0.0031	98.33 ± 0.75	98.43 ± 0.65	99.88 ± 0.11
MultiHash U	0.0875 ± 0.0087	86.54 ± 1.91	81.91 ± 2.84	92.89 ± 1.35
MultiHash P	0.0816 ± 0.0096	87.71 ± 1.62	83.13 ± 3.03	93.75 ± 1.25
TrigK U	0.0127 ± 0.0018	98.25 ± 0.84	98.06 ± 0.69	99.89 ± 0.04
TrigK P	0.0106 ± 0.0018	98.26 ± 0.63	98.29 ± 0.49	99.90 ± 0.05

Arm	Native BCE	Precision %	Recall %	Average precision %
TrigHash U	0.0237 ± 0.0035	96.02 ± 1.31	96.59 ± 1.29	99.54 ± 0.18
TrigHash P	0.0206 ± 0.0040	96.45 ± 0.75	96.55 ± 2.01	99.57 ± 0.21
TrigG121 U	0.0095 ± 0.0024	98.03 ± 1.10	98.05 ± 0.52	99.90 ± 0.05
TrigG121 P	0.0090 ± 0.0027	98.20 ± 1.18	98.12 ± 0.72	99.92 ± 0.04

For spiral TrigK_P, mean native BCE is 0.250964 compared with clipped BCE 0.240354. Both belong in the interpretation. ECE uses the frozen ten-bin rule and Brier uses raw probabilities; calibration diagnostics are not interchangeable with discrimination or classification performance. All reliability-bin counts, predictions and extreme-logit diagnostics remain in the endpoint evidence.

Appendix B Every primary comparison

The interval for each contrast has 99.75% nominal confidence, implementing the registered twenty-comparison Bonferroni family at 5%. The independent unit is the mean of two paired seed differences for one population: n = 8 and seven degrees of freedom. Small-sample t assumptions limit nominal coverage. The 0.01 BCE reference is a practical benchmark, not a power calculation or equivalence margin.

C1 is MultiK P minus U; C2 is TrigK P minus U. C3 is the MultiK priority effect minus the MultiHash priority effect; C4 is the equivalent TrigK minus TrigHash interaction. C5 is MultiK_P minus TrigG121_P. Negative C1 or C2 favors priority. Negative C3 or C4 means a more favorable priority effect under geometry, without by itself proving either absolute effect beneficial. Positive C5 favors the global control.

Checkerboard

Contrast	Mean BCE difference	Nominal interval	Interpretation
C1	-0.00351	[-0.01346, +0.00644]	Inconclusive
C2	-0.00926	[-0.01140, -0.00712]	Negative direction
C3	-0.00450	[-0.01498, +0.00598]	Inconclusive
C4	-0.00802	[-0.01343, -0.00261]	Negative direction
C5	+0.45654	[+0.42031, +0.49277]	Positive direction

All five registered checkerboard contrasts. No negative interval is wholly below −0.01.

Population	C1	C2	C3	C4	C5
1	+0.00056	-0.01008	-0.00123	-0.00947	+0.47774
2	-0.00318	-0.00832	-0.00554	-0.00832	+0.44140
3	-0.00641	-0.00961	-0.00642	-0.00961	+0.42938
4	-0.00193	-0.01006	-0.00193	-0.01006	+0.49824
5	+0.00062	-0.01084	-0.00004	-0.01084	+0.44734
6	-0.01124	-0.00756	-0.01278	-0.00373	+0.44212
7	+0.00563	-0.01024	+0.00545	-0.01024	+0.45414
8	-0.01215	-0.00739	-0.01354	-0.00190	+0.46196

All eight paired population differences, in registered population order 01 to 08. Values are rounded for display only.

Spiral

Contrast	Mean BCE difference	Nominal interval	Interpretation
C1	-0.00508	[-0.01203, +0.00187]	Inconclusive

Contrast	Mean BCE difference	Nominal interval	Interpretation
C2	-0.00866	[-0.01146, -0.00586]	Negative direction
C3	-0.00579	[-0.01299, +0.00142]	Inconclusive
C4	-0.00678	[-0.01225, -0.00131]	Negative direction
C5	+0.30249	[+0.22495, +0.38003]	Positive direction

All five registered spiral contrasts. No negative interval is wholly below −0.01.

Population	C1	C2	C3	C4	C5
1	-0.00631	-0.00869	-0.00762	-0.00757	+0.22436
2	+0.00065	-0.00750	+0.00090	-0.00199	+0.35279
3	-0.00103	-0.00859	-0.00664	-0.00925	+0.31944
4	-0.00899	-0.00540	-0.00744	-0.00177	+0.23120
5	-0.00972	-0.00902	-0.01119	-0.00709	+0.33679
6	-0.00653	-0.00862	-0.00528	-0.00614	+0.32040
7	-0.00866	-0.01041	-0.00968	-0.00960	+0.31780
8	-0.00002	-0.01103	+0.00066	-0.01081	+0.31711

All eight paired population differences, in registered population order 01 to 08. Values are rounded for display only.

Noisy annulus

Contrast	Mean BCE difference	Nominal interval	Interpretation
C1	-0.00168	[-0.00378, +0.00042]	Inconclusive
C2	-0.00045	[-0.00404, +0.00315]	Inconclusive
C3	-0.00400	[-0.01046, +0.00246]	Inconclusive
C4	-0.00025	[-0.00523, +0.00474]	Inconclusive
C5	-0.00573	[-0.01438, +0.00291]	Inconclusive

All five registered noisy annulus contrasts. No negative interval is wholly below −0.01.

Population	C1	C2	C3	C4	C5
1	-0.00170	-0.00315	-0.00510	-0.00292	-0.00370
2	-0.00062	-0.00011	-0.00024	+0.00174	-0.00473
3	-0.00200	+0.00070	-0.00748	+0.00070	-0.00638
4	-0.00243	-0.00438	-0.00846	-0.00565	-0.00450
5	-0.00165	+0.00174	-0.00579	+0.00219	-0.00969

Population	C1	C2	C3	C4	C5
6	-0.00419	-0.00021	-0.00654	+0.00283	-0.00511
7	+0.00002	+0.00000	+0.00270	-0.00260	+0.00347
8	-0.00084	+0.00181	-0.00104	+0.00172	-0.01524

All eight paired population differences, in registered population order 01 to 08. Values are rounded for display only.

Rare disks

Contrast	Mean BCE difference	Nominal interval	Interpretation
C1	-0.00409	[-0.00604, -0.00214]	Negative direction
C2	-0.00213	[-0.00298, -0.00127]	Negative direction
C3	+0.00187	[-0.00051, +0.00426]	Inconclusive
C4	+0.00095	[-0.00028, +0.00219]	Inconclusive
C5	+0.00273	[-0.00116, +0.00662]	Inconclusive

All five registered rare disks contrasts. No negative interval is wholly below −0.01.

Population	C1	C2	C3	C4	C5
1	-0.00290	-0.00198	+0.00001	+0.00004	+0.00203
2	-0.00230	-0.00266	+0.00362	+0.00155	+0.00389
3	-0.00345	-0.00288	+0.00349	+0.00145	+0.00237
4	-0.00456	-0.00133	+0.00095	+0.00105	+0.00065
5	-0.00432	-0.00207	+0.00099	+0.00073	+0.00756
6	-0.00413	-0.00191	+0.00317	+0.00208	-0.00046
7	-0.00490	-0.00165	+0.00242	+0.00091	+0.00331
8	-0.00612	-0.00252	+0.00035	-0.00017	+0.00250

All eight paired population differences, in registered population order 01 to 08. Values are rounded for display only.

Across the twenty contrasts, six intervals support a negative direction, two support a positive direction, and twelve cross zero. Four of the six favorable intervals are direct priority effects; two are trigonometric routing interactions. The two positive intervals are C5 on checkerboard and spiral. Counting these directions is a summary of the registered family, not twenty separate uncorrected discoveries.

Appendix C Learning progress

The following threshold table shows reached fits out of 16 and the median first observed update among those fits. “0 of 16; censored” means no fit reached the target by 15,000 updates. All nonreachers are right-censored at that horizon. Metrics are recorded every 75 updates; a median can lie between two observed updates. These nested fits do not become 16 independent populations.

Checkerboard

Arm	90% reached and median	95% reached and median	97% reached and median
MultiK U	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiK P	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiHash U	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiHash P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigK U	16 of 16; 11625	0 of 16; censored	0 of 16; censored
TrigK P	16 of 16; 10350	0 of 16; censored	0 of 16; censored
TrigHash U	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigHash P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigG121 U	16 of 16; 300	16 of 16; 937.5	16 of 16; 2700
TrigG121 P	16 of 16; 300	16 of 16; 862.5	16 of 16; 2325

Checkerboard fixed learning rounds

Each table cell is mean full-training macro-F1 / mean validation macro-F1, both in percent. The arithmetic mean uses all eight stored population blocks; each block already averages the two paired seeds. Rounds 25, 50, 100 and 200 were fixed in advance and correspond to 1,875, 3,750, 7,500 and 15,000 optimizer updates. These descriptive observations use the parameter state at each fixed round, not the validation-selected checkpoint. They are not test learning curves, primary intervals, extra independent replications or a new selection rule.

Arm	Round 25 Train / Val %	Round 50 Train / Val %	Round 100 Train / Val %	Round 200 Train / Val %
MultiK U	69.65 / 68.77	72.56 / 71.63	74.57 / 73.73	76.29 / 75.26
MultiK P	70.04 / 68.94	72.57 / 71.67	74.58 / 73.42	76.35 / 75.75
MultiHash U	54.26 / 50.12	54.68 / 50.20	55.29 / 50.34	55.37 / 49.93
MultiHash P	54.17 / 49.95	54.81 / 50.06	55.19 / 50.20	55.18 / 49.83
TrigK U	83.49 / 83.05	86.14 / 85.65	88.66 / 88.02	90.90 / 90.37
TrigK P	83.70 / 83.17	86.30 / 85.67	88.93 / 88.46	91.51 / 90.88

Arm	Round 25 Train / Val %	Round 50 Train / Val %	Round 100 Train / Val %	Round 200 Train / Val %
TrigHash U	56.54 / 50.62	56.32 / 51.06	56.42 / 50.25	56.16 / 49.98
TrigHash P	56.35 / 50.54	56.70 / 50.85	56.49 / 50.89	56.45 / 50.87
TrigG121 U	97.11 / 96.38	98.02 / 97.19	98.64 / 97.80	98.96 / 98.18
TrigG121 P	97.40 / 96.35	98.20 / 97.53	98.90 / 98.02	99.33 / 98.35

All ten arms retained in registered order. U = uniform replacement sampling; P = prioritized sampling. Complete eight-population records: learning_inventory_v01/COMPANION_EVIDENCE.json, fixed_learning_groups. Authoritative source: committed capsule priority_summary.exposure_and_learning.

Spiral

Arm	90% reached and median	95% reached and median	97% reached and median
MultiK U	3 of 16; 13650	0 of 16; censored	0 of 16; censored
MultiK P	3 of 16; 11325	0 of 16; censored	0 of 16; censored
MultiHash U	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiHash P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigK U	16 of 16; 4425	0 of 16; censored	0 of 16; censored
TrigK P	16 of 16; 4162.5	2 of 16; 12825	0 of 16; censored
TrigHash U	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigHash P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigG121 U	16 of 16; 75	16 of 16; 375	9 of 16; 4500
TrigG121 P	16 of 16; 75	16 of 16; 375	12 of 16; 4500

Spiral fixed learning rounds

Each table cell is mean full-training macro-F1 / mean validation macro-F1, both in percent. The arithmetic mean uses all eight stored population blocks; each block already averages the two paired seeds. Rounds 25, 50, 100 and 200 were fixed in advance and correspond to 1,875, 3,750, 7,500 and 15,000 optimizer updates. These descriptive observations use the parameter state at each fixed round, not the validation-selected checkpoint. They are not test learning curves, primary intervals, extra independent replications or a new selection rule.

Arm	Round 25 Train / Val %	Round 50 Train / Val %	Round 100 Train / Val %	Round 200 Train / Val %
MultiK U	81.45 / 80.66	83.62 / 82.93	84.93 / 84.41	86.14 / 85.59
MultiK P	81.14 / 80.33	83.71 / 82.82	85.30 / 84.40	86.78 / 86.14
MultiHash U	57.02 / 54.56	57.94 / 55.81	59.27 / 56.77	59.29 / 56.63

Arm	Round 25 Train / Val %	Round 50 Train / Val %	Round 100 Train / Val %	Round 200 Train / Val %
MultiHash P	56.90 / 54.71	58.31 / 55.82	59.59 / 56.57	59.43 / 56.64
TrigK U	88.76 / 88.14	90.15 / 89.64	91.82 / 91.32	93.28 / 92.64
TrigK P	88.71 / 88.21	90.45 / 90.09	92.35 / 91.93	93.56 / 92.94
TrigHash U	61.20 / 57.89	61.10 / 57.75	61.72 / 58.10	61.30 / 57.93
TrigHash P	61.58 / 58.05	61.32 / 57.96	61.51 / 58.32	61.66 / 58.02
TrigG121 U	96.98 / 96.00	97.51 / 96.36	97.69 / 96.55	98.00 / 96.71
TrigG121 P	97.21 / 96.25	97.58 / 96.48	97.97 / 96.66	98.27 / 96.87

All ten arms retained in registered order. U = uniform replacement sampling; P = prioritized sampling. Complete eight-population records: learning_inventory_v01/COMPANION_EVIDENCE.json, fixed_learning_groups. Authoritative source: committed capsule priority_summary.exposure_and_learning.

Noisy annulus

Arm	90% reached and median	95% reached and median	97% reached and median
MultiK U	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiK P	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiHash U	0 of 16; censored	0 of 16; censored	0 of 16; censored
MultiHash P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigK U	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigK P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigHash U	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigHash P	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigG121 U	0 of 16; censored	0 of 16; censored	0 of 16; censored
TrigG121 P	0 of 16; censored	0 of 16; censored	0 of 16; censored

Noisy annulus fixed learning rounds

Each table cell is mean full-training macro-F1 / mean validation macro-F1, both in percent. The arithmetic mean uses all eight stored population blocks; each block already averages the two paired seeds. Rounds 25, 50, 100 and 200 were fixed in advance and correspond to 1,875, 3,750, 7,500 and 15,000 optimizer updates. These descriptive observations use the parameter state at each fixed round, not the validation-selected checkpoint. They are not test learning curves, primary intervals, extra independent replications or a new selection rule.

Arm	Round 25 Train / Val %	Round 50 Train / Val %	Round 100 Train / Val %	Round 200 Train / Val %
MultiK U	87.42 / 86.80	87.50 / 87.00	87.50 / 86.88	87.79 / 87.31
MultiK P	87.31 / 86.82	87.63 / 86.98	87.65 / 86.96	87.83 / 87.12
MultiHash U	77.66 / 75.79	78.54 / 76.80	78.98 / 77.12	79.13 / 77.53
MultiHash P	77.70 / 75.82	78.36 / 76.67	79.01 / 77.27	79.15 / 77.16
TrigK U	88.27 / 87.69	88.25 / 87.59	88.24 / 87.43	88.44 / 87.70
TrigK P	88.26 / 87.48	88.32 / 87.74	88.41 / 87.54	88.45 / 87.72
TrigHash U	83.37 / 81.67	83.53 / 81.83	83.58 / 81.87	83.54 / 81.87
TrigHash P	83.38 / 81.74	83.54 / 81.88	83.61 / 82.04	83.58 / 81.79
TrigG121 U	88.02 / 87.22	87.90 / 87.13	87.84 / 86.91	87.81 / 87.08
TrigG121 P	88.02 / 87.14	88.04 / 87.23	88.03 / 87.20	88.06 / 87.25

All ten arms retained in registered order. U = uniform replacement sampling; P = prioritized sampling. Complete eight-population records: learning_inventory_v01/COMPANION_EVIDENCE.json, fixed_learning_groups. Authoritative source: committed capsule priority_summary.exposure_and_learning.

Rare disks

Arm	90% reached and median	95% reached and median	97% reached and median
MultiK U	16 of 16; 375	16 of 16; 750	16 of 16; 1650
MultiK P	16 of 16; 375	16 of 16; 750	16 of 16; 1425
MultiHash U	16 of 16; 6750	0 of 16; censored	0 of 16; censored
MultiHash P	16 of 16; 4275	0 of 16; censored	0 of 16; censored
TrigK U	16 of 16; 75	16 of 16; 262.5	16 of 16; 487.5
TrigK P	16 of 16; 75	16 of 16; 262.5	16 of 16; 487.5
TrigHash U	16 of 16; 150	16 of 16; 525	16 of 16; 1387.5
TrigHash P	16 of 16; 150	16 of 16; 525	16 of 16; 1162.5
TrigG121 U	16 of 16; 75	16 of 16; 150	16 of 16; 150
TrigG121 P	16 of 16; 75	16 of 16; 150	16 of 16; 150

Rare disks fixed learning rounds

Each table cell is mean full-training macro-F1 / mean validation macro-F1, both in percent. The arithmetic mean uses all eight stored population blocks; each block already averages the two paired seeds. Rounds 25, 50, 100 and 200 were fixed in advance and correspond to 1,875, 3,750, 7,500 and 15,000 optimizer updates. These descriptive observations use the parameter state at each fixed

round, not the validation-selected checkpoint. They are not test learning curves, primary intervals, extra independent replications or a new selection rule.

Arm	Round 25 Train / Val %	Round 50 Train / Val %	Round 100 Train / Val %	Round 200 Train / Val %
MultiK U	97.34 / 96.84	98.05 / 97.70	98.51 / 98.23	98.72 / 98.45
MultiK P	97.86 / 97.35	98.65 / 98.26	99.07 / 98.66	99.37 / 98.95
MultiHash U	86.32 / 84.42	89.35 / 87.76	91.41 / 89.61	92.36 / 90.41
MultiHash P	86.69 / 84.66	90.49 / 88.92	92.47 / 90.48	93.16 / 91.09
TrigK U	98.74 / 98.37	99.08 / 98.67	99.29 / 98.96	99.52 / 99.11
TrigK P	98.83 / 98.46	99.30 / 98.87	99.59 / 99.13	99.76 / 99.24
TrigHash U	98.09 / 96.92	98.76 / 97.19	99.16 / 97.54	99.50 / 97.84
TrigHash P	98.34 / 97.03	99.00 / 97.58	99.46 / 97.71	99.79 / 97.80
TrigG121 U	99.43 / 98.84	99.56 / 99.01	99.73 / 99.02	99.85 / 99.00
TrigG121 P	99.61 / 99.03	99.81 / 99.15	99.92 / 99.06	99.96 / 99.04

All ten arms retained in registered order. U = uniform replacement sampling; P = prioritized sampling. Complete eight-population records: learning_inventory_v01/COMPANION_EVIDENCE.json, fixed_learning_groups. Authoritative source: committed capsule priority_summary.exposure_and_learning.

All 16 checkerboard fits per TrigK arm reach 90%, with median 11,625 updates for U and 10,350 for P; none reach 95%. All checkerboard global fits reach 97%, with medians 2,700 and 2,325. No annulus arm reaches 90%. Early disk crossings within the first 750 uniform updates cannot be attributed to priority. All fits still completed 15,000 updates, so these observations do not establish early-stopping speed savings.

Appendix D Exposure and diagnostic inventory

Effective record count is the reciprocal of the sum of squared actual sampling probabilities for the actual sampling distribution. It measures concentration, not additional independent data. Uniform sampling has an effective count of 4,800 but visits only about 3,034 distinct records per round because draws are with replacement. The table retains every arm and task. Maximum visits is the population-mean endpoint maximum, not a new overall extremum.

Checkerboard

Arm	Effective records	Unique per round	Maximum visits	Flipped draws %
MultiK U	4800.0	3034.1	255.2	0.00
MultiK P	4626.6	3003.1	404.8	0.00
MultiHash U	4800.0	3034.1	255.2	0.00
MultiHash P	4786.9	3032.0	265.6	0.00
TrigK U	4800.0	3034.1	255.2	0.00
TrigK P	4352.1	2952.4	544.2	0.00
TrigHash U	4800.0	3034.1	255.2	0.00
TrigHash P	4774.7	3030.7	275.9	0.00
TrigG121 U	4800.0	3034.1	255.2	0.00
TrigG121 P	3185.1	2744.9	887.0	0.00

Spiral

Arm	Effective records	Unique per round	Maximum visits	Flipped draws %
MultiK U	4800.0	3034.5	255.4	0.00
MultiK P	4418.0	2967.1	529.3	0.00
MultiHash U	4800.0	3034.5	255.4	0.00
MultiHash P	4751.6	3026.0	297.0	0.00
TrigK U	4800.0	3034.5	255.4	0.00
TrigK P	4157.9	2921.9	695.0	0.00
TrigHash U	4800.0	3034.5	255.4	0.00
TrigHash P	4722.7	3021.1	309.1	0.00
TrigG121 U	4800.0	3034.5	255.4	0.00
TrigG121 P	2889.0	2713.3	1541.5	0.00

Noisy annulus

Arm	Effective records	Unique per round	Maximum visits	Flipped draws %
MultiK U	4800.0	3034.5	252.9	9.92
MultiK P	4235.7	2942.2	515.6	18.46
MultiHash U	4800.0	3034.5	252.9	9.92
MultiHash P	4486.4	2980.0	476.2	14.09
TrigK U	4800.0	3034.5	252.9	9.92
TrigK P	4190.8	2935.5	534.4	19.05
TrigHash U	4800.0	3034.5	252.9	9.92
TrigHash P	4335.5	2955.5	534.6	16.28
TrigG121 U	4800.0	3034.5	252.9	9.92
TrigG121 P	4144.5	2927.9	571.2	19.30

Rare disks

Arm	Effective records	Unique per round	Maximum visits	Flipped draws %
MultiK U	4800.0	3035.0	252.7	0.00
MultiK P	2714.4	2769.0	1724.5	0.00
MultiHash U	4800.0	3035.0	252.7	0.00
MultiHash P	3121.6	2757.2	1296.5	0.00
TrigK U	4800.0	3035.0	252.7	0.00
TrigK P	2834.8	2797.8	1732.8	0.00
TrigHash U	4800.0	3035.0	252.7	0.00
TrigHash P	3022.0	2776.8	1324.0	0.00
TrigG121 U	4800.0	3035.0	252.7	0.00
TrigG121 P	3493.2	2912.7	1777.5	0.00

The complete 200-round train and validation curves, fixed rounds 25, 50, 100 and 200, native losses, high-loss sampling fractions and all per-round exposure traces remain in the official endpoint metrics and the priority summary. No test checkpoint curves were inspected. The companion learning tables preserve the eight population blocks instead of treating technical draws as independent replications.

Complete diagnostic slice inventory

The inventory includes all 560 task/arm/policy/bin summaries and all 4,480 population blocks. It contains 0 empty blocks and 30 observed single-class blocks (30 negative-only, 0 positive-only).

Composition is repeated across ten model arms: there are 448 unique population/policy/bin data slices, with 0 empty and 3 single-class unique slices. These are overlapping descriptive partitions, not independent replications or extra test records.

Counts below include all ten arms and all eight populations for the stated task and policy. Count and prevalence ranges describe individual population blocks, not pooled averages. No slice is ranked or selected by its performance. Each task/arm is described using five hash-defined slices, five kmeans-defined slices and four geometric-margin bins; these diagnostic partitions do not imply that every model has routed banks. In a single-class slice, the frozen two-class macro-F1 convention assigns zero F1 to the absent class; the maximum macro-F1 is therefore 0.5 even for perfect predictions of the present class. Interpret such slice scores alongside class composition and BCE.

Task	Policy	Slices / blocks	Empty	Single class	Records	Positive %
Checkerboard	Hash	50 / 400	0	0	245–386	44.41–59.38
Checkerboard	Kmeans	50 / 400	0	0	223–414	43.63–59.49
Checkerboard	Margin	40 / 320	0	0	141–841	44.99–58.49
Spiral	Hash	50 / 400	0	0	237–408	44.83–54.69
Spiral	Kmeans	50 / 400	0	0	237–401	42.81–57.24
Spiral	Margin	40 / 320	0	0	149–834	36.26–57.88
Noisy annulus	Hash	50 / 400	0	0	221–411	42.01–52.92
Noisy annulus	Kmeans	50 / 400	0	0	231–400	9.52–60.68
Noisy annulus	Margin	40 / 320	0	0	145–824	36.25–66.34
Rare disks	Hash	50 / 400	0	0	249–440	7.84–14.48
Rare disks	Kmeans	50 / 400	0	20	229–400	0.00–23.40
Rare disks	Margin	40 / 320	0	10	157–819	0.00–21.66

Complete, unfiltered 560-slice JSON and every eight-population block: learning_inventory_v01/COMPANION_EVIDENCE.json, diagnostic_population_slices. Per-slice inventory: diagnostic_inventory.per_slice. Authoritative committed source: analysis/official_results/stage_B_v01/diagnostic_summary.json. No diagnostic significance intervals are introduced.

Appendix E Complete classical inference

Latency units are microseconds, written µs: 1,000 µs equals 1 ms. Each value times one complete call for the stated batch of 1, 32 or 1,024 records, not a per-record latency. All three sessions are retained. The following tables show 48 circuit task-arm-batch groups with their paired ratios, savings and construction-only break-even; an additional table preserves the ratio-of-medians metric. Thus all 144 independent paired metric summaries remain visible, separately from the 48 derived break-even summaries.

Checkerboard inference latency

Cells are complete-call latency in microseconds, including raw-input routing/feature/core/readout operations with resident model data. Fifteen trials collapse to a session median, then three session medians collapse within each model; two seed values average within a population. The displayed value is the median across all eight populations. Full population values and ranges are retained. Technical repeats do not increase independent sample size. For learned Trig arms, original denotes the equally optimized same-checkpoint table kernel.

Arm	Representation	Batch 1 µs	Batch 32 µs	Batch 1024 µs
MultiK U	Original	321.03	1477.42	2283.83
MultiK U	Compiled	71.62	91.62	446.53
MultiK P	Original	315.75	1492.80	2309.28
MultiK P	Compiled	70.33	92.48	443.68
MultiHash U	Original	322.48	1523.05	3224.08
MultiHash U	Compiled	75.58	126.88	1350.42
MultiHash P	Original	320.28	1520.20	3235.88
MultiHash P	Compiled	74.58	126.90	1360.25
TrigK U	Original	61.85	70.12	347.43
TrigK P	Original	63.25	70.45	355.23
TrigHash U	Original	64.90	99.62	1285.83
TrigHash P	Original	64.65	98.75	1286.98
TrigG121 U	Original	43.25	55.35	929.02
TrigG121 P	Original	42.77	54.58	896.00

Checkerboard matched circuit savings

Arm	Batch	Paired ratio	Saving µs	Build break even	Defined pops
MultiK U	1	4.54×	245.75	7.87	8/8

Arm	Batch	Paired ratio	Saving µs	Build break even	Defined pops
MultiK U	32	15.91×	1363.78	1.40	8/8
MultiK U	1024	5.22×	1794.10	1.05	8/8
MultiK P	1	4.50×	241.95	5.52	8/8
MultiK P	32	15.89×	1371.98	0.95	8/8
MultiK P	1024	5.18×	1846.53	0.70	8/8
MultiHash U	1	4.38×	244.55	5.43	8/8
MultiHash U	32	12.12×	1382.95	0.99	8/8
MultiHash U	1024	2.35×	1812.30	0.72	8/8
MultiHash P	1	4.35×	243.70	9.78	8/8
MultiHash P	32	11.90×	1371.12	1.69	8/8
MultiHash P	1024	2.35×	1834.92	1.28	8/8

Ratios and savings are paired at trial/model level before session, seed and population aggregation. They are not division or subtraction of the aggregate latency cells above. A ratio above one favors the compiled call. Construction-only break-even divides that model's single build time by its retained positive matched saving after the three-session median; both seeds must have positive savings for a population mean to be defined. Undefined cases are retained and counted. Units are complete-call equivalents: round upward to obtain whole calls. Loading, additional router/storage setup costs, hardware cost and changes in future execution conditions are excluded. No accuracy gain or physical quantum speedup is implied.

Spiral inference latency

Cells are complete-call latency in microseconds, including raw-input routing/feature/core/readout operations with resident model data. Fifteen trials collapse to a session median, then three session medians collapse within each model; two seed values average within a population. The displayed value is the median across all eight populations. Full population values and ranges are retained. Technical repeats do not increase independent sample size. For learned Trig arms, original denotes the equally optimized same-checkpoint table kernel.

Arm	Representation	Batch 1 µs	Batch 32 µs	Batch 1024 µs
MultiK U	Original	315.88	1490.20	2255.68
MultiK U	Compiled	70.58	93.15	452.75
MultiK P	Original	317.85	1460.50	2251.60
MultiK P	Compiled	69.38	96.10	442.15
MultiHash U	Original	317.28	1517.33	3275.22
MultiHash U	Compiled	71.73	128.45	1336.12

Arm	Representation	Batch 1 µs	Batch 32 µs	Batch 1024 µs
MultiHash P	Original	322.18	1543.60	3290.72
MultiHash P	Compiled	73.83	131.43	1364.98
TrigK U	Original	61.10	69.50	356.07
TrigK P	Original	60.70	69.58	347.40
TrigHash U	Original	66.40	100.28	1283.08
TrigHash P	Original	64.60	99.75	1281.45
TrigG121 U	Original	43.20	54.60	911.88
TrigG121 P	Original	42.75	54.38	920.08

Spiral matched circuit savings

Arm	Batch	Paired ratio	Saving µs	Build break even	Defined pops
MultiK U	1	4.53×	244.38	5.03	8/8
MultiK U	32	16.03×	1386.70	0.88	8/8
MultiK U	1024	5.17×	1795.88	0.71	8/8
MultiK P	1	4.59×	241.18	4.81	8/8
MultiK P	32	15.98×	1360.33	0.86	8/8
MultiK P	1024	5.25×	1783.35	0.66	8/8
MultiHash U	1	4.38×	242.65	5.03	8/8
MultiHash U	32	11.91×	1361.92	0.89	8/8
MultiHash U	1024	2.34×	1838.70	0.68	8/8
MultiHash P	1	4.37×	243.18	4.67	8/8
MultiHash P	32	12.27×	1407.55	0.80	8/8
MultiHash P	1024	2.35×	1834.78	0.63	8/8

Ratios and savings are paired at trial/model level before session, seed and population aggregation. They are not division or subtraction of the aggregate latency cells above. A ratio above one favors the compiled call. Construction-only break-even divides that model's single build time by its retained positive matched saving after the three-session median; both seeds must have positive savings for a population mean to be defined. Undefined cases are retained and counted. Units are complete-call equivalents: round upward to obtain whole calls. Loading, additional router/storage setup costs, hardware cost and changes in future execution conditions are excluded. No accuracy gain or physical quantum speedup is implied.

Noisy annulus inference latency

Cells are complete-call latency in microseconds, including raw-input routing/feature/core/readout operations with resident model data. Fifteen trials collapse to a session median, then three session medians collapse within each model; two seed values average within a population. The displayed value is the median across all eight populations. Full population values and ranges are retained. Technical repeats do not increase independent sample size. For learned Trig arms, original denotes the equally optimized same-checkpoint table kernel.

Arm	Representation	Batch 1 µs	Batch 32 µs	Batch 1024 µs
MultiK U	Original	320.23	1480.08	2285.08
MultiK U	Compiled	70.85	96.85	436.80
MultiK P	Original	317.73	1475.58	2231.30
MultiK P	Compiled	70.33	93.00	439.35
MultiHash U	Original	320.78	1530.45	3264.33
MultiHash U	Compiled	75.73	128.07	1374.90
MultiHash P	Original	325.45	1552.25	3225.85
MultiHash P	Compiled	74.80	129.53	1358.92
TrigK U	Original	61.33	69.50	345.85
TrigK P	Original	61.25	70.00	350.73
TrigHash U	Original	64.25	100.00	1252.67
TrigHash P	Original	65.65	99.90	1259.60
TrigG121 U	Original	42.83	54.48	911.80
TrigG121 P	Original	43.05	54.68	913.00

Noisy annulus matched circuit savings

Arm	Batch	Paired ratio	Saving µs	Build break even	Defined pops
MultiK U	1	4.54×	244.90	6.76	8/8
MultiK U	32	15.71×	1376.42	1.18	8/8
MultiK U	1024	5.21×	1777.80	0.92	8/8
MultiK P	1	4.54×	247.30	4.87	8/8
MultiK P	32	15.91×	1368.10	0.85	8/8
MultiK P	1024	5.16×	1770.03	0.67	8/8
MultiHash U	1	4.33×	242.45	4.65	8/8

Arm	Batch	Paired ratio	Saving µs	Build break even	Defined pops
MultiHash U	32	12.13×	1392.17	0.82	8/8
MultiHash U	1024	2.33×	1798.90	0.67	8/8
MultiHash P	1	4.36×	244.28	4.78	8/8
MultiHash P	32	12.00×	1383.80	0.83	8/8
MultiHash P	1024	2.34×	1831.62	0.63	8/8

Ratios and savings are paired at trial/model level before session, seed and population aggregation. They are not division or subtraction of the aggregate latency cells above. A ratio above one favors the compiled call. Construction-only break-even divides that model's single build time by its retained positive matched saving after the three-session median; both seeds must have positive savings for a population mean to be defined. Undefined cases are retained and counted. Units are complete-call equivalents: round upward to obtain whole calls. Loading, additional router/storage setup costs, hardware cost and changes in future execution conditions are excluded. No accuracy gain or physical quantum speedup is implied.

Rare disks inference latency

Cells are complete-call latency in microseconds, including raw-input routing/feature/core/readout operations with resident model data. Fifteen trials collapse to a session median, then three session medians collapse within each model; two seed values average within a population. The displayed value is the median across all eight populations. Full population values and ranges are retained. Technical repeats do not increase independent sample size. For learned Trig arms, original denotes the equally optimized same-checkpoint table kernel.

Arm	Representation	Batch 1 µs	Batch 32 µs	Batch 1024 µs
MultiK U	Original	313.53	1469.60	2224.60
MultiK U	Compiled	70.58	96.12	420.98
MultiK P	Original	313.03	1473.05	2237.50
MultiK P	Compiled	70.12	91.60	425.95
MultiHash U	Original	323.53	1556.25	3241.08
MultiHash U	Compiled	73.53	133.90	1335.45
MultiHash P	Original	321.98	1494.17	3281.47
MultiHash P	Compiled	74.25	126.98	1321.70
TrigK U	Original	61.60	69.00	341.75
TrigK P	Original	61.27	69.17	347.80
TrigHash U	Original	65.03	99.23	1246.03
TrigHash P	Original	65.10	98.65	1255.05

Arm	Representation	Batch 1 µs	Batch 32 µs	Batch 1024 µs
TrigG121 U	Original	43.50	54.50	893.05
TrigG121 P	Original	42.73	54.18	894.43

Rare disks matched circuit savings

Arm	Batch	Paired ratio	Saving µs	Build break even	Defined pops
MultiK U	1	4.59×	239.93	6.49	8/8
MultiK U	32	15.88×	1364.17	1.12	8/8
MultiK U	1024	5.33×	1783.88	0.86	8/8
MultiK P	1	4.55×	240.33	4.78	8/8
MultiK P	32	15.96×	1362.75	0.85	8/8
MultiK P	1024	5.36×	1789.50	0.64	8/8
MultiHash U	1	4.39×	247.95	4.79	8/8
MultiHash U	32	11.82×	1400.85	0.84	8/8
MultiHash U	1024	2.35×	1793.05	0.63	8/8
MultiHash P	1	4.42×	243.85	5.84	8/8
MultiHash P	32	12.15×	1350.73	1.06	8/8
MultiHash P	1024	2.39×	1910.50	0.73	8/8

Ratios and savings are paired at trial/model level before session, seed and population aggregation. They are not division or subtraction of the aggregate latency cells above. A ratio above one favors the compiled call. Construction-only break-even divides that model's single build time by its retained positive matched saving after the three-session median; both seeds must have positive savings for a population mean to be defined. Undefined cases are retained and counted. Units are complete-call equivalents: round upward to obtain whole calls. Loading, additional router/storage setup costs, hardware cost and changes in future execution conditions are excluded. No accuracy gain or physical quantum speedup is implied.

All three timing sessions and drift

Task	S2 / S1	S3 / S1	Max / min session
Checkerboard	1.002 [0.302–2.439]	0.589 [0.190–0.703]	1.788 [1.440–5.268]
Spiral	0.993 [0.413–3.430]	0.585 [0.265–1.045]	1.787 [1.045–5.636]
Noisy annulus	0.992 [0.394–2.185]	0.590 [0.264–0.677]	1.754 [1.481–3.792]
Rare disks	0.990 [0.446–2.051]	0.587 [0.295–0.717]	1.775 [1.394–3.388]

Each cell is the median [full minimum–maximum] of paired condition ratios across all 672 task conditions, including all arms, representations and batch sizes. These descriptive drift ratios compare sessions of the same workloads; they are not extra model comparisons. All three sessions remain in every primary timing summary. OS scheduling, caches, frequency scaling and background activity are not experimentally controlled.

Each session contains 640 selected models, 256 additional compiled circuit representations, 2,688 conditions and 40,320 timed calls. The independent audit checks all 120,960 raw timed rows across three sessions. The 16,128 warmups are committed counters supported by the producer loop; individual warmup call times are not retained or independently observed. Each timed trial contains one complete raw-input call. Memory was sampled at start/end and at all 32 population blocks.

Complete retained ratio of medians

These 48 values are the separately retained ratio-of-medians timing metric, aggregated through the same model, paired-seed and population hierarchy. They are copied from the independently checked timing summary, not calculated by dividing the displayed aggregate latency cells. They complement the 48 paired trial ratios and 48 matched savings above; break-even is a separate construction-cost diagnostic.

Checkerboard

Arm	Batch	Ratio of medians
MultiHash P	1	4.386×
MultiHash P	32	12.182×
MultiHash P	1024	2.348×
MultiHash U	1	4.452×
MultiHash U	32	12.397×
MultiHash U	1024	2.361×
MultiK P	1	4.556×
MultiK P	32	16.015×
MultiK P	1024	5.274×
MultiK U	1	4.564×
MultiK U	32	16.438×
MultiK U	1024	5.239×

Spiral

Arm	Batch	Ratio of medians
MultiHash P	1	4.376×
MultiHash P	32	12.430×

Arm	Batch	Ratio of medians
MultiHash P	1024	2.394×
MultiHash U	1	4.440×
MultiHash U	32	12.189×
MultiHash U	1024	2.395×
MultiK P	1	4.586×
MultiK P	32	16.046×
MultiK P	1024	5.192×
MultiK U	1	4.571×
MultiK U	32	16.240×
MultiK U	1024	5.209×

Noisy annulus

Arm	Batch	Ratio of medians
MultiHash P	1	4.416×
MultiHash P	32	12.248×
MultiHash P	1024	2.354×
MultiHash U	1	4.395×
MultiHash U	32	12.481×
MultiHash U	1024	2.360×
MultiK P	1	4.579×
MultiK P	32	16.108×
MultiK P	1024	5.232×
MultiK U	1	4.577×
MultiK U	32	15.664×
MultiK U	1024	5.254×

Rare disks

Arm	Batch	Ratio of medians
MultiHash P	1	4.445×
MultiHash P	32	12.040×
MultiHash P	1024	2.427×

Arm	Batch	Ratio of medians
MultiHash U	1	4.399×
MultiHash U	32	12.010×
MultiHash U	1024	2.356×
MultiK P	1	4.593×
MultiK P	32	16.465×
MultiK P	1024	5.454×
MultiK U	1	4.609×
MultiK U	32	16.047×
MultiK U	1024	5.376×

Appendix F Costs and execution record

The table records mean full-training elapsed time, summed process CPU over 16 fits, and mean nested refresh and sampler CPU. Training elapsed includes validation and evidence writing but ends before final serialization; process CPU has a different measurement boundary. Nested counters are included in broader work and must not be added to it. These concurrent training observations are not isolated inference benchmarks.

Checkerboard

Arm	Mean elapsed s	Total CPU s	Mean refresh CPU s	Mean sampler CPU s
MultiK U	398.17	6469.38	0.903	0.132
MultiK P	398.13	6470.05	0.910	0.155
MultiHash U	397.92	6467.56	0.905	0.156
MultiHash P	397.87	6462.61	0.920	0.137
TrigK U	11.45	162.16	0.343	0.112
TrigK P	11.38	163.31	0.337	0.130
TrigHash U	11.35	159.59	0.354	0.123
TrigHash P	11.09	159.75	0.344	0.122
TrigG121 U	9.52	132.83	0.806	0.157
TrigG121 P	9.62	133.16	0.806	0.155

Spiral

Arm	Mean elapsed s	Total CPU s	Mean refresh CPU s	Mean sampler CPU s
MultiK U	398.80	6472.86	0.916	0.139
MultiK P	399.18	6480.20	0.896	0.156
MultiHash U	399.36	6480.64	0.873	0.158
MultiHash P	399.43	6484.44	0.896	0.141
TrigK U	11.62	165.12	0.338	0.102
TrigK P	11.75	163.92	0.370	0.113
TrigHash U	11.45	161.59	0.370	0.124
TrigHash P	11.51	163.00	0.383	0.130
TrigG121 U	9.67	134.45	0.825	0.176
TrigG121 P	9.52	132.88	0.786	0.201

Noisy annulus

Arm	Mean elapsed s	Total CPU s	Mean refresh CPU s	Mean sampler CPU s
MultiK U	399.21	6476.92	0.906	0.151
MultiK P	399.36	6473.94	0.920	0.145
MultiHash U	399.73	6478.94	0.901	0.151
MultiHash P	399.00	6468.86	0.871	0.146
TrigK U	11.44	162.48	0.352	0.117
TrigK P	11.62	161.86	0.348	0.108
TrigHash U	11.40	162.53	0.370	0.122
TrigHash P	11.45	161.50	0.416	0.123
TrigG121 U	9.57	132.91	0.834	0.183
TrigG121 P	9.59	132.23	0.805	0.169

Rare disks

Arm	Mean elapsed s	Total CPU s	Mean refresh CPU s	Mean sampler CPU s
MultiK U	400.05	6487.75	0.886	0.149
MultiK P	399.60	6484.94	0.865	0.169
MultiHash U	399.82	6491.22	0.829	0.137
MultiHash P	395.49	6417.75	0.878	0.144

Arm	Mean elapsed s	Total CPU s	Mean refresh CPU s	Mean sampler CPU s
TrigK U	11.41	161.66	0.351	0.124
TrigK P	11.60	162.70	0.365	0.107
TrigHash U	11.34	162.66	0.349	0.123
TrigHash P	11.49	162.91	0.346	0.104
TrigG121 U	9.52	131.88	0.820	0.184
TrigG121 P	9.43	131.66	0.858	0.150

Registered trace-array payload per full fit is 72,529,600 bytes for a 70-parameter circuit, 79,129,600 bytes for a 125-parameter routed trigonometric model, and 78,649,600 bytes for the 121-parameter global model. The 720 official full fits comprise 288, 288 and 144 of these models, respectively, giving a calculated 55,003,392,000 bytes of trace-array payload. This excludes engineering and reproduction traces, other arrays, file containers and package overhead; it is not total measured disk use. Detailed audit traces should not be mistaken for model size or minimal deployment footprint. Cache construction, validation, diagnostic passes and audit work are separate from compact predictor coefficients.

Uniform controls are intentionally padded with full diagnostic refreshes. These timings cannot quantify the native end-to-end penalty of adding priority to an uninstrumented uniform trainer. Pure gradient computation, filesystem I/O, energy use and thermal behavior were not separately measured. ETEL's 64 logical bookkeeping slots are not 64 workers or an observed speedup; at most five CPU workers ran complete fits.

Derived phase costs and accounting boundaries

Resource availability in the following table follows the inner phase-summary records. The independent timing audit does not retain whole-process resource counters in its inner summary, hence the two Unrecorded cells. Its separate external completion records 270.890625 seconds of process CPU and a maximum live-sampled peak working-set observation of 173,367,296 bytes. Those broader counters are reported again in the operations record below. The memory observation is sampled, not a guaranteed final process peak. Nested counters overlap and must not be added.

Phase	Inner wall s	Inner CPU s	Process CPU s	Peak RAM MiB
Clean reproductions	505.70	497.22	780.06	201.93
Independent reproduction	2.21	2.12	286.17	207.24
Conversion	19.57	12.88	302.53	107.91
Independent conversion	99.76	98.61	396.75	108.88
Timing session 1	48.98	46.67	630.62	143.89
Timing session 2	48.23	46.30	612.62	143.87
Timing session 3	27.00	25.41	573.36	143.92

Phase	Inner wall s	Inner CPU s	Process CPU s	Peak RAM MiB
Independent timing audit	5.80	5.69	Unrecorded	Unrecorded

Inner counters start after phase release verification and may exclude imports, source/hash validation and output preflight. Whole-process CPU and peak working set are shown only where explicitly recorded through the sampling point. Unrecorded is not zero. Peak RAM is process working set, not device memory or memory allocated solely to one model. These nested counters must not be added to each other. External invocation wall times belong to the separate operations supplement and must not be added to inner wall time. Construction and component probes are single instrumented observations that perturb execution.

Arm	Model and checkpoint µs	Representation verify and load µs	Samples / models
MultiK U	1702.60	1123.60	192 / 64
MultiK P	1630.65	1083.85	192 / 64
MultiHash U	1692.05	1121.85	192 / 64
MultiHash P	1681.45	1079.55	192 / 64
TrigK U	1588.00	1058.05	192 / 64
TrigK P	1586.05	1064.40	192 / 64
TrigHash U	1542.75	1036.00	192 / 64
TrigHash P	1536.75	1024.65	192 / 64
TrigG121 U	1597.80	1064.95	192 / 64
TrigG121 P	1514.30	1035.00	192 / 64

Medians across 192 single instrumented observations per arm: 64 models across four tasks, each observed in all three sessions. These are loading diagnostics, not primary resident-call latencies, and their sum is not a measured cold-start latency. All model-specific loading and component records, session identities, memory samples and full scopes are retained in DERIVED_EVIDENCE.json.

External invocation costs and retained events

Completed phase	Launch to exit wall s
Official scoring	416.101
Independent score and statistics replay	391.220
Four clean same-environment reproductions	813.629
Independent clean-reproduction comparison	311.398
ReportLab figure render v02	1.531
ReportLab figure render v03	1.061
Classical conversion and learned Trig repacking	336.442

Completed phase	Launch to exit wall s
Independent conversion comparison	432.625
Isolated timing session 1	685.925
Isolated timing session 2	666.154
Isolated timing session 3	624.270
Independent timing evidence audit	295.366

These external wall measurements include the launch and exit boundary. They overlap the narrower inner timers; do not add them. The derived-phase table reports separately captured process CPU and inner timers. The scoring/replay and presentation records below are additional operations, not inference speed or a complete study cost total.

Operation	Inner wall s	Inner CPU s	Process CPU s
Official scoring	112.661	107.453	381.344
Independent score and statistics replay	279.598	252.219	Unrecorded

Independent replay has no retained whole-process CPU or peak-working-set observation. Its inner 252.219 s CPU is not a substitute. The independent clean-reproduction 2.207 s comparison excludes preflight; its 286.172 s whole-process CPU and 311.398 s outer wall capture broader work. Missing observations are never treated as zero.

Brief syntheticcheck overlap during development

A 16-test synthetic suite overlapped development training for 1.672 s wall / 1.03125 s CPU before the coordinated hold arrived. Training was not rerun. Training wall is concurrent operating cost, not isolated latency.

Metadata helper correction

A source-inventory list/dictionary mismatch caused a metadata helper TypeError before producing a review. Corrected binding traversal; no numerical computation or scientific source change. Runtime cost not recorded.

Synthetic preparation leasepurpose rejection

An unsupported lease purpose was rejected before numerical fixtures. Failed outer command wall 0.7344731 s; CPU not recorded. Only preparation-purpose strings were corrected; fresh approved fixtures passed. Frozen training sources unchanged.

Independent B audit prelaunch PID reuse

First independent B invocation exited1 before numerical audit after 68.9682307 s outer wall; process CPU not retained. Windows reused completed training PID5920 for a protected service. After the service exited naturally, unchanged native checks of all760 current identities passed. Root authorized a fresh same-source full audit with a distinct log. No API bypass, fallback wrapper, source edit, ledger repair, replacement seed or training rerun occurred. Attempt02 then passed all640 endpoints with exit0. Successful lease-owning wall1557.8193071000278s and process CPU1528.25s; outer launch-to-exit wall1558.0421744s. These nested scopes are not additive.

Author B audit timing scope

Reported phase wall 2683.4936157999327 s starts inside audit() before evidence verification and ends before final receipt. Actual interpreter CPU 2759.078125 s includes imports/lease checks; peak working set197861376 bytes. Neither is launcher end-to-end wall. Detached exit code was not retained; completion relies on PASS receipt, full stdout terminator, empty stderr, owner exit and unchanged bindings. The immutable helper inherited fit-child wording; describe this as audit-process CPU.

Independent audit completion summarizer correction

A metadata-only helper tried to sum per-endpoint probe lists as counts and raised TypeError. Corrected to count each existing list length; all7680 probe records retained. No numerical audit, training or outcome computation repeated. Runtime cost not recorded.

Matplotlib unavailable in authoring runtime

Initial figure rendering failed at the optional import. Retained inner0.0042053000007015467s wall/0.0s CPU excludes prior input verification. No package installed and no scientific environment changed; separate ReportLab/SVG/Sharp sources produced fresh versions.

SVG boldfont serialization correction

The first ReportLab render preserved all numbers but used a bold alias that fell back to serif. A new helper changed only SVG font serialization; finalv03 figures passed visual review. Both attempts and costs remain retained.

Learning supplement capsulehash argument typo

An extra substring in the supplied SHA was rejected before any output directory was created. The unchanged helper then used the exact approved SHA and succeeded. Failed outer/CPU cost was not recorded; missing is not zero. No model or study outcome changed.

Assembly metadata inspection corrections

Two read-only inspection commands failed before output writes: default Windows decoding could not read UTF8; corrected invocation then assumed a block key type instead of kind. Explicit UTF8 and existing kind schema were used for the successful inspection. No experiment, source or result was changed; elapsed tool observations0.294s/0.303s are command waits, not measured process costs.

Successful independent timing exit counters

The successful independent timing audit's external completion retained a separate process-exit measurement scope. Actual child PID 41112: CPU 270.891 s, maximum live-sampled peak working-set observation 173,367,296 bytes; recorded resource-query error None. These counters include broader preflight and audit work, are separate from any summary-sampled counters, and must not be added to overlapping inner CPU or wall time.

Timing session one postexit metadata path correction

The timing process exited successfully. Its first metadata verifier assumed scientific freeze keys were relative to the study root, but they are relative to quantum/. It failed before completing metadata acceptance. Only the verifier base path changed in a new v02 file; all timing output and original source remained unchanged, with no repeated timing session.

Independent timing preaudit PID reuse

The first audit invocation failed closed before numerical timing replay because a completed attempt PID was reused and the current process denied native access. It created no scientific audit outputs. A first reviewed coordination adapter covered that one exact original identity, but the next invocation encountered a second reused protected PID and also failed before numerical replay. The final separately reviewed adapter validates all 760 individually enumerated accepted closure proofs and still calls the unchanged native API first. Only access denied may use an exact proof that the original creation identity had ended. The entrypoint changed; original numerical/statistical source and inputs did not. Both failures and final provenance remain retained. No timing session was rerun. The first failed external invocation took 14.018 s; actual auditor CPU and peak memory were unrecorded.

Second independent timing preaudit identity failure

Unchanged native API returned access denied on another exact historical creation identity (PID48176); the first versioned amendment only authorized the separate PID49660 case and correctly failed closed. This external timing_v02 attempt created no scientific audit output and ran no numerical audit. The successful separately authorized external attempt is timing_v03; its retained scientific output directory remains timing_v01. The second failed outer invocation took 97.394 s. Actual child PID 40264 used 89.031 s CPU; peak memory was unrecorded.

Timing coordination amendment and bounded fixtures

The adapter records its own source and phase-specific authority for the complete exact 760-identity inventory. Native identity/source/endpoint checks and the same exclusive kernel lock remain; no ledger writes, service changes, source replacement or monkeypatch occurred. Eighteen stdlib fixtures tested the final amendment without model or timing calls. The earlier 14-fixture candidate and initial 13-test observation are also preserved.

Readonly alternate command observation

The bounded read-only command returned the retained status FAIL_CLOSED. It did not start scientific auditing or establish a successful native identity observation.

Readonly alternate command observation

No process or service metadata was returned by these queries. This does not establish current process identity, death, or a process-specific protection mechanism.

Operations presentation helper binding correction

The first report-accounting helper invocation verified a bound Python source but then incorrectly attempted to parse it as JSON. It exited before creating the supplement. A separate v07 helper keeps every byte-identity check and uses JSON parsing only when consuming JSON data. No scientific phase, inference timing or numeric result was rerun. The failed presentation invocation took 0.231429 s outer wall and 0.09375 s CPU; peak memory was unrecorded.

Every retained event and exact source receipt is listed in report_work/final_report/OPERATIONS_COST_EVIDENCE_SUPPLEMENT_v03.json. Preparation, presentation and process-identity failures are distinguished from scientific training failures; all 760 official fits completed.

Appendix G Evidence and preservation

The root after-study preservation receipt matches the before record for all 53,427 v0.7 payloads totalling 6,533,819,005 bytes and all four Baseline7 seal files, including the archive-hash agreement. This statement is bound to those completed receipts; report assembly does not repeat their filesystem scan.

The four full clean-directory retrains reuse the preserved scientific Python 3.12.14, NumPy 2.5.4 and SciPy 1.18.1 environment with BLAS limited to one thread. They regenerate data and routers and copy frozen scientific sources, without copying trained checkpoints or rebuilding dependencies. This establishes the registered same-environment checks; it does not establish arbitrary relocation or reproduction on different hardware.

Evidence	Package relative location
Registered design and streams	governance/STUDY_PLAN.json and RUN_MATRIX.json
Training and data sources	quantum/ and governance/source freezes
Data provenance and exclusion	quantum/data/DATA_SEAL.json and governance/DATA_REGISTRATION.json
All attempts and checkpoints	execution/state/ledger.sqlite3 and each retained attempt metadata/endpoint directory
Official endpoint predictions and diagnostics	analysis/official_results/stage_B_v01/
Independent predictions and complete replay	analysis/independent_results/stage_B_v01/
All fixed learning and diagnostic populations	report_work/final_report/learning_inventory_v01/COMPANION_EVIDENCE.json
Clean reproduction and independent comparison	analysis/reproduction_results/full_v01/ and analysis/independent_reproduction_results/full_v01/
Conversion and all timing result locators	report_work/final_report/derived_blocks_v01/DERIVED_EVIDENCE.json and COMMIT.json
Local reproduction procedure and limits	analysis/reproduction/PACKAGE_REPLAY_GUIDE.txt
Final package verification contract	analysis/independent_package/README_FINAL_VERIFICATION.txt
Preservation and report QA	governance/PRIOR_PRESERVATION_BEFORE.json and PRIOR_PRESERVATION_AFTER.json; report_work/final_report/qa/

Read the retained reproduction guide before running commands: original absolute provenance is preserved, and archival verification does not silently authorize new training. Final delivery acceptance is recorded separately in root and independent verification receipts; this report is not the delivery receipt.

Full hashes, exact provenance paths and the original review targets are retained in this content's source bindings and assembly requirement summary. Evidence locators supplement the complete numeric tables; they do not replace a required arm, contrast or population difference.

Report edition and rights

Study v0.8. Report edition 1 prepared on 29 October 2026.

Anthony Siljeg — Founder, Alnotrea Labs · Company affiliation: Alnotrea Pty Ltd · Q-Pathformer Initiative. Anthony is the licensor; company affiliation is not an ownership transfer.

All original report paragraphs, tables, figures, numerical findings, limitations and historical recommendations are retained without alteration. Original internal labels identify the historical source edition. This appended note and document metadata are the only editorial changes. No scientific result was recalculated or removed.

Historical local paths, host configuration, execution and usage accounting identify retained evidence. The full archive remains separately preserved under C:\Q2. This curated release does not include raw datasets, trained checkpoints, installed environments or full operational archives. Retained archive sizes are not download sizes or hardware minimums.

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